

WEEK #2

Introduction and Software Requirements for Java Programs

Problem 1. What are the softwares that help to run Java programs?

The main software required for developing and running Java programs is the Java Development Kit (JDK). The important components/software used for Java programming are:

- JDK (Java Development Kit) – Used to develop, compile, and run Java programs.
- JRE (Java Runtime Environment) – Provides the environment required to run Java applications.
- JVM (Java Virtual Machine) – Executes Java bytecode.
- IDE (Integrated Development Environment) – Provides an environment for writing, compiling, debugging, and running Java programs.

Examples of Java IDEs include Eclipse, NetBeans, and IntelliJ IDEA.

A Java source file normally has the .java extension. The Java compiler converts the source code into bytecode, which is stored in a .class file. The JVM then executes this bytecode.

Result: Java programming software was identified successfully. The JDK is required for Java development, while the JVM executes Java bytecode.

Problem 2. What is JDK and JRE?

JDK – Java Development Kit

JDK stands for Java Development Kit. It is a software development kit used to develop Java applications. It contains tools required for Java programming, including the Java compiler.

Important tools include:

- javac – Java compiler
- java – Used to launch/run Java applications
- javadoc – Generates documentation
- jar – Creates and manages Java archive files

The basic process is: Java Source Code → javac → Bytecode → JVM → Output

JRE – Java Runtime Environment

JRE stands for Java Runtime Environment. It provides the environment required to run Java applications. It includes the JVM, Java class libraries, and other supporting files required for running Java applications.

JDK	JRE	Main Difference
Java Development Kit	Java Runtime Environment	JDK is for development; JRE is for running applications.
Contains development tools	Provides runtime environment	JDK contains tools such as the compiler.
Includes runtime components	Contains JVM and runtime libraries	JRE does not provide the Java compiler.

Result: The difference between JDK and JRE was studied successfully. JDK is mainly used for development, whereas JRE provides the environment for running Java applications.

Problem 3. What is Eclipse IDE?

Eclipse IDE is an Integrated Development Environment used for software development, including Java programming. IDE stands for Integrated Development Environment.

Eclipse provides several facilities for Java programmers:

- Writing Java source code.
- Syntax highlighting.
- Automatic code suggestions.
- Compiling Java programs.
- Running Java programs.
- Finding and displaying programming errors.
- Debugging programs.
- Managing Java projects and files.

The general workflow is: Create Project → Create Class → Write Code → Compile → Run → View Output

Result: Eclipse IDE was studied successfully. It provides an integrated environment for writing, compiling, debugging, and executing Java programs.

Problem 4. How to run a Java program in Eclipse/NetBeans IDE?

A. Running a Java Program in Eclipse

1. Install a suitable JDK on the computer.
2. Launch Eclipse IDE.
3. Select File → New → Java Project and enter a project name such as JavaLab.

4. Right-click the src folder and select New → Class. Enter the class name HelloWorld.
5. Write the Java program shown below.
6. Right-click the Java file and select Run As → Java Application, or click the Run button.
7. The output appears in the Console window.

Program:

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello World");  
    }  
}
```

Expected Output:

Hello World

B. Running a Java Program in NetBeans

8. Install a suitable JDK on the computer.
9. Launch NetBeans IDE.
10. Select File → New Project and create the appropriate Java project.
11. Create a Java class named HelloWorld.
12. Write the same HelloWorld program.
13. Right-click the project and select Run, or use the Run button.
14. The output is displayed in the Output window.

Expected Output:

Hello World

Result: The Java program was successfully created and executed using an IDE. The program displayed: Hello World.